Operating Systems
Week 14
Lecture/Discussion 14

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Previous

- Individual Assignment: Understanding Volatile and Non-Volatile Memory

Today Agenda

- Exploring the structure of an operating system's I/O subsystem

Flow I/O Components

User/App

↓

System Call Interface

↓

I/O Scheduler

↓

Device Driver

↓

Device Controller

↓

I/O Device

↑

← Interrupt / Acknowledgement

Activity

OS I/O Subsystem – We Are the System!

Objective is to visualize the structure and function of the I/O subsystem by acting out the roles of its components — such as devices, device drivers, device controllers, and the I/O management layer.

Subsystem Components

Application / User
System Call Interface
I/O Scheduler
Device Driver
Device Controller
I/O Device (e.g., Keyboard, Printer, Disk)

I/O Subsystem components activity

E.g., *“User wants to print a document”* or *“User wants to read a file from disk.”*
Use this scenario to walk through the I/O subsystem:

1. **User/Application**: Makes a request (e.g., print "Hello").
2. **System Call Interface**: Converts the request to an OS-level I/O command.
3. **I/O Scheduler**: Decides the order of device access if multiple requests exist.
4. **Device Driver**: Converts commands into device-specific operations.
5. **Device Controller**: Sends/receives signals to/from the physical device.
6. **I/O Device**: Executes the operation (e.g., print text).

I/O Subsystem components activity

data flow by passing a paper with the "command" written on it.

Interrupts

Have the device “signal” back when the job is done — simulate an **interrupt** sent to the OS, then the **OS notifies** the application.
E.g., “Printer finished printing!”

Brief Investigation

Question	Expected Understanding
What does the device driver do?	Translates OS instructions to device commands
Why is the I/O scheduler important?	Manages device access efficiently
What role does the controller play?	Interface between hardware and software
How is data sent back to the application?	Via interrupts or polling